



Sri Chaitanya IIT Academy., India.

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Regional Office: Chennai - Tamilnadu

PERIODIC TABLE_CHEMICAL BONDING DAY-01 (15-03-24)

SECTION – I

(ONE OR MORE CORRECT ANSWER TYPE)

- Which of the following statements is/are correct?
 - Among Li^- , Be^- , B^- and C^- , the least stable ion is Be^- .
 - Electro negativity of 'H' and 'N' are 2.1 and 3.0. Then percent ionic character of H – N bond is 17.24.
 - First ionization of energy $Cu > Ag < Au$.
 - Electron gain enthalpy of Br is negative where as that of Xe is positive.
- Select the incorrect order of periodic properties of species.
 - $Ga < Al$: atomic radius.
 - $Cu > Zn$: metallic radius.
 - $P < S$: Second ionization energy.
 - $He > Ne$: Positive value of $\Delta_{eg}H$.
- Which of the following statements is/are correct?

Statement I:

Hybridisation of carbon atom	S - character	Electronegativity
sp	50%	2.48
sp ²	33.30%	2.75

sp^3	25%	3.25
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Statement II: Electron gain enthalpy of $O > S$

Statement III: Hydration energy of $H^+ > Li^+ > Na^+ > K^+$.

A) I, II, III are correct

B) II, III are correct

C) III only is correct

D) I, III are correct

4. Correct match of the following

A) Uus – a symbol of element with atomic no. 106

B) Ge and As – metalloids of 4th period

C) Common valencies of 16th group elements are 2 and 6

D) IE of $X^{2+}(g)$ is equal to $\Delta_{eg}H$ of $X^+(g)$

5. A non - metallic third period element “A” whose spin-only magnetic moment is $\sqrt{15}$ B.M combines with element “B” to give a compound $A_m B_n$. “B” is the second most electronegative element in the periodic table. Then, which of the following statements are correct?

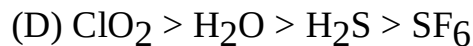
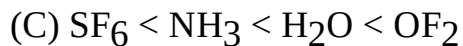
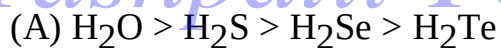
A) If mass percentage of ‘B’ in the compound is 56.3, that gives the formula A_2B_5

B) If mass percentage of ‘B’ in the compound is 43.6, that gives the the formula A_2B_3

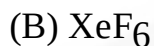
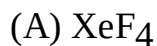
C) Acidic nature of $A_2B_5 > A_2B_3$

D) ${}_{29}Cu^{59}$ is the isodiaphere of A^{31}

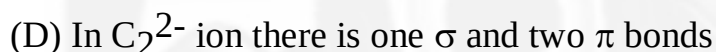
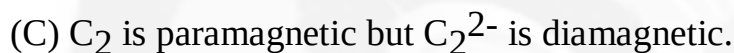
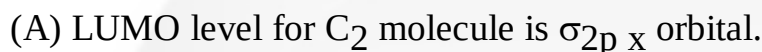
6. Correct order about bond angle is :



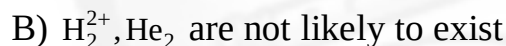
7. Which of the following are polar ?



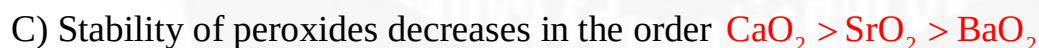
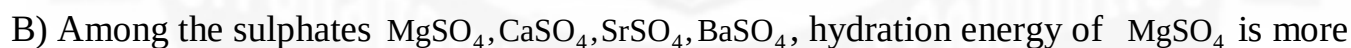
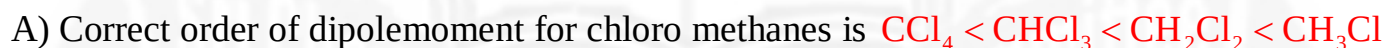
8. According to Molecular orbital theory which of the following statement is correct?
(Assume x as the bonding axis)



9. Identify correct statements



10. Which of the following statements are correct



SECTION – II

(INTEGER ANSWER TYPE)

11. Consider the orders,

a) Electron affinity $F > Cl > Br > I$

b) Size: $Li^+(aq) < K^+(aq) < Rb^+(aq)$

c) Acidic strength of $N_2O_3 > P_2O_3 > As_2O_3$ d) Size of $La^{+3} < Lu^{+3}$

e) Number of electrons in the penultimate shell of Cu-atom $<$ Zn – atom

f) Hydration energy of $F^- > Cl^- > Br^- > I^-$

if 'a' and 'b' are number of correct and incorrect statements, then $|a - b|^2$ is

12. Total number of transition elements that are present in 5th period in modern periodic table among the following is:

Pm, Sm, Rb, Sr, Po, At, Ag

13. Outer Electronic configuration of *I is $5s^x 5p^y$. Then, $(x + y)$ is*

14. In how many of the following elements highest valency “(According to the group no)” is not possible in any one of their compounds.

Al, Si, P, S, C, N, O, F, Xe

15. Pauling electro negativity of carbon, oxygen and fluorine are x,y and z respectively.

$\frac{(x + y) - z}{2}$ is equal to .

16. The bond order of the underlined species; NOHSO₄ is :

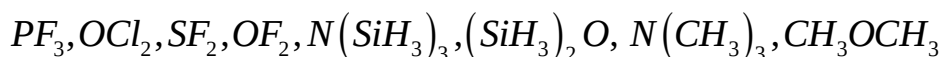
17. If the dipole moment of AB molecule is given by 1.2 D and A-B the bond length is 1Å. Then the ratio of the percentage covalent character of the bond to the percentage ionic character is?

18. How many of the following are isoelectronic and having same geometry as that of SO_4^{2-} ,

$S_2O_3^{2-}, ClO_4^-, PO_4^{3-}, SiF_4, SiO_4^{4-}, SO_3^{2-}, SF_4, NH_4^+$

19. The number of 90° interaction between the bond pairs in PCl_5 is

20. In how many of the following compounds the bond angles are more than expected due to delocalization of lone pair electrons according to VSEPR theory



SECTION-III

(PARAGRAPH QUESTIONS)

Paragraph for Question Nos. 21 to 22

The f-block elements are known as inner transition elements because they involve the filling of inner sub shells 4f (or) 5f. Because lanthanum closely resembles lanthanides, it is usually included in any discussion of lanthanides.

Ba	La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu	Hf
Ra	Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr	Rf

There is fairly decrease in the sizes with increasing atomic number. The Continuous decrease in radii is known as Lanthanides Contraction. Lanthanides exhibit the oxidation state of +3. Some of them also exhibit the oxidation state of +2 and +4. And in such case species present in +4 oxidation state acts as good oxidizing agent.

21. Out of the following, which one is strongest base?
- A) $Eu(OH)_3$ B) $La(OH)_3$ C) $Lu(OH)_3$ D) $Dy(OH)_3$
22. Out of the following, which one is strongest oxidising agent in aqueous medium?
- A) Yb^{2+} B) Sm^{2+} C) Pm^{3+} D) Ce^{4+}

Paragraph for Question Nos. 23 to 24

The percentage ionic character of a single bond may be estimated from the ratio of the observed dipole moment to the calculated moments, assuming oppositely charged ions located at distance equal to the bond length

23. The observed dipole moment of HBr is 0.79D. calculate the percentage ionic character of the bond in HBr if bond distance is $1.42 \text{ \AA}$:

- a) 16.9% b) 12.0% c) 5.0% d) 20%

24. Dichloro benzene has three possible position isomers ortho, meta and para. Arrange the isomers in increasing order of dipole moment

- a) ortho < meta < para b) meta < ortho < para
c) para < meta < ortho d) para < ortho < meta

Paragraph for Question Nos. 25 to 26

One hydrogen bonding represented as $X-H \cdots D$ denotes the interaction between the species D and the acceptor species X-H through the hydrogen end. H atom should remain the atoms (i.e., X and D) of more electro negative non-metallic elements.

25. Which of the following compound does not have hydrogen bonding?

- a) K_2HPO_4 b) K_2HPO_3 c) chloralhydrate d) H_2O

26. The strength of H-bonding order is:

- a) $NH_3 > H_2O > H-F$ b) $H_2O > NH_3 > HF$
c) $HF > H_2O > NH_3$ d) $HF == H_2O == NH_3$

SECTION - IV

(Matching List Type)

27. Match the following Column-I with Column – II

Column – I

Column – II

(Compound)

(Hybridization of the central atom)

a) SO_3

i) sp

b) I_3^-

ii) sp^2

c) CO_2

iii) sp^3

d) PO_4^{3-}

iv) sp^3d

(a) a-ii, b-iii, c-i, d-iv (b) a-i, b-iii, c-ii, d-iv

(c) a-iv, b-ii, c-iii, d-I (d) a-ii, b-iv, c-i d-iii

28. Match the following list – I with List – II

List – I

List – II

a) $XeO_3, XeF_2, ClF_3, PCl_5$

i) All are sp^3d except one

b) $CH_4, XeO_3, ClO_4^-, POCl_3$

ii) All are sp^3 except one

c) $BF_4^-, SO_4^{2-}, CO_3^{2-}, POCl_3$

iii) All are sp^3

d) $SO_2, SnCl_2, SO_3, SO_2Cl_2$

iv) All are sp^2 except one

(a) a-i,, b-iv, c-iii, d-ii

(b) a-i, b-iii, c-ii, d-iv

(c) a-iv, b-ii, c-iii, d-i

(d) a-ii, b-iv, c-ii d-iii

29.

Column I		Column II	
a.	C-H bond in ethyne	i.	sp-s overlap
b.	P-Cl bond in $POCl_3$	ii.	sp^3 d-p overlap
c.	Br-Br bond in Br_3^-	iii.	sp^3 -p overlap
d.	C-C bond in ethane	iv.	sp^3 - sp^3 overlap

The correct code is

(a) a-i, b-ii, c-iv, d-iii

(b) a-i, b-iii, c-ii, d-iv

(c) a-iv, b-ii, c-iii, d-I

(d) a-ii, b-iv, c-i, d-iii

30.

Column I (Molecule/ion)		Column II (bond angle)	
a.	CO_3^{2-}	i.	180°
b.	ICl_2^-	ii.	$190^\circ 28'$
c.	XeF_6	iii.	120°
d.	BF_4^-	iv.	90° (approximately)

The correct code is

(a) a-iii, b-ii, c-iv, d-i

(b) a-ii, b-iii, c-iv, d-i

(c) a-i, b-iv, c-iii, d-ii

(d) a-iii, b-i, c-iv, d-ii

KEY SHEET

CHEMISTRY:

1	ABCD	2	BD	3	C	4	BC	5	ABCD	6	ABD
7	BC	8	ABD	9	ABCD	10	ABD	11	4	12	3
13	7	14	2	15	1	16	3	17	3	18	5
19	6	20	4	21	B	22	D	23	B	24	C
25	B	26	C	27	D	28	B	29	B	30	D